



Software Engineering Final Lab Documentation

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Project Idea

The Smart Parking System is a desktop application developed using C# Windows Forms and SQLite that addresses the challenge of urban parking management. The system enables drivers to view real-time availability of parking slots and reserve them before arriving at the facility. It supports two user roles: Drivers, who can search, book, and manage their reservations with a live countdown timer; and Administrators, who can add, update, or remove parking slots and monitor all bookings through a centralized control panel. Each booking is assigned an automatic expiry time based on the selected duration, and the system releases the slot automatically once the booking period ends.



Problem Statement

Finding a parking space in crowded urban areas is a persistent and frustrating challenge for drivers. Traditional parking systems are entirely manual, forcing drivers to physically search for available spots, which leads to unnecessary fuel consumption, increased traffic congestion, and wasted time. There is no mechanism to check slot availability in advance, no way to reserve a spot before arrival, and no automated system to free slots once the booking duration expires. This inefficiency not only affects individual drivers but also reduces the overall throughput and revenue potential of parking facilities, as administrators have no centralized visibility into real-time occupancy or booking history.

Problem Solution

The Smart Parking System solves the above problems through the following key features:

1. **Real-Time Slot Visibility:** Drivers can view all available parking slots at any time with live status updates.
2. **Advance Reservation:** Drivers can reserve a slot before arriving and receive an instant booking confirmation with the exact expiry time.
3. **Automatic Slot Release:** When a booking expires, the system automatically marks the slot as available without any manual intervention.
4. **Live Countdown Timer:** The My Bookings view shows a real-time countdown (e.g., 57m 15s left) so drivers always know how much time remains.
5. **Manual Cancellation:** Drivers can cancel active bookings at any time, immediately freeing the slot for other users.
6. **Booking History:** Drivers can view a complete history of all their past and active bookings with timestamps and amounts.
7. **Admin Management Panel:** Administrators can add, update, and delete parking slots; view all driver bookings; view registered users; and monitor real-time statistics.
8. **Role-Based Access Control:** The system enforces separate dashboards and permissions for Drivers and Administrators, ensuring data security.

System Analysis

Use Case Diagram

The Use Case Diagram illustrates the interactions between the two main actors (Driver and Admin) and the Smart Parking System. The Driver can perform: User Registration, User Login, View Available Parking Slots, Book Parking Slot (which includes View Available Parking Slots via an include relationship), Cancel Booking, and View Booking History. The Admin can perform: User Login, Add Parking Slot, Update Parking Slot, Delete Parking Slot, and View All Bookings. All user operations require prior login.

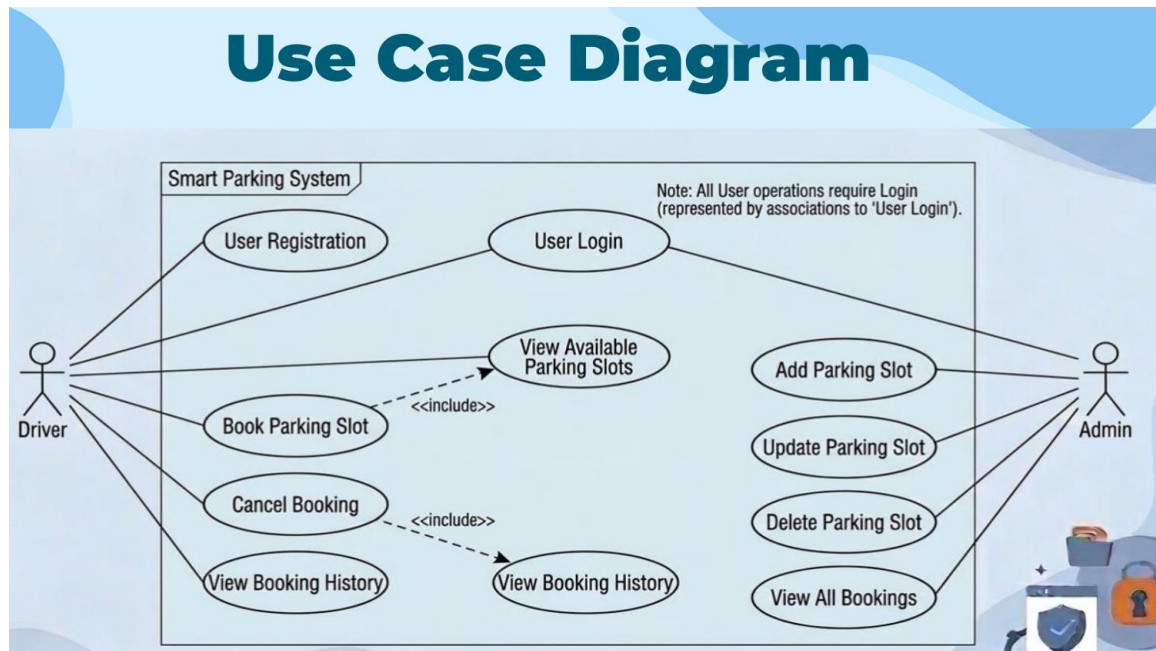


Figure 1: Use Case Diagram

Class Diagram

The Class Diagram models the static structure of the system. The base class User contains userID, password, and email, with login(), logout(), and getProfile() methods. Driver extends User and adds licenseNumber with register(), updateProfile(), and viewBookingHistory() methods, and has a one-to-many (1 to 0..*) association with Booking. Admin extends User and adds adminCode with manageUsers(), generateReports(), and configureSystem() methods. ParkingSlot holds slotID, status, location, and pricePerHour. Booking holds bookingID, startTime, duration, totalAmount, and status with a (0..1 to 1) association with ParkingSlot.

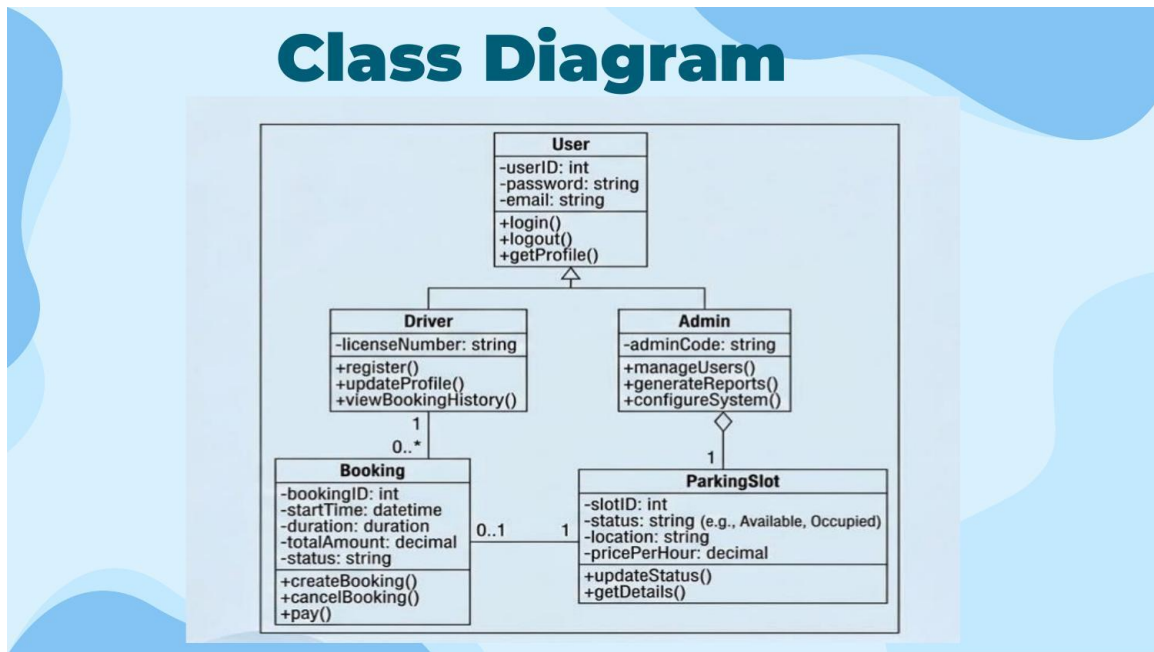


Figure 2: Class Diagram

Sequence Diagram - Driver

The Driver Sequence Diagram shows the booking flow: the Driver requests available slots, the system queries the database and displays them. The Driver selects a slot, the system checks availability, and the Driver confirms with a duration. The system processes the booking, stores the EndTime, updates the slot status to Booked, saves to the database, and displays confirmation.

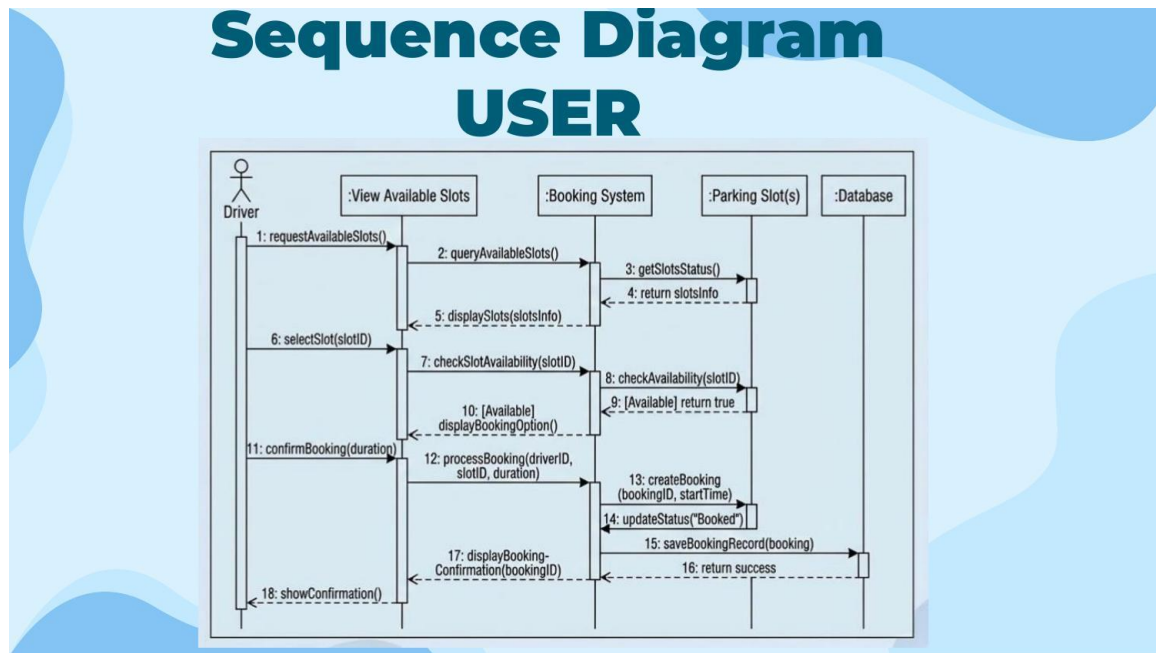


Figure 3: Sequence Diagram - Driver Booking Flow

Sequence Diagram - Admin

The Admin Sequence Diagram shows the administrative flow for creating and managing parking slots. The Admin creates slots with their locations and configures pricing details. The admin can also view all slots, check availability, and manage bookings following the same database interaction pattern as the driver flow.

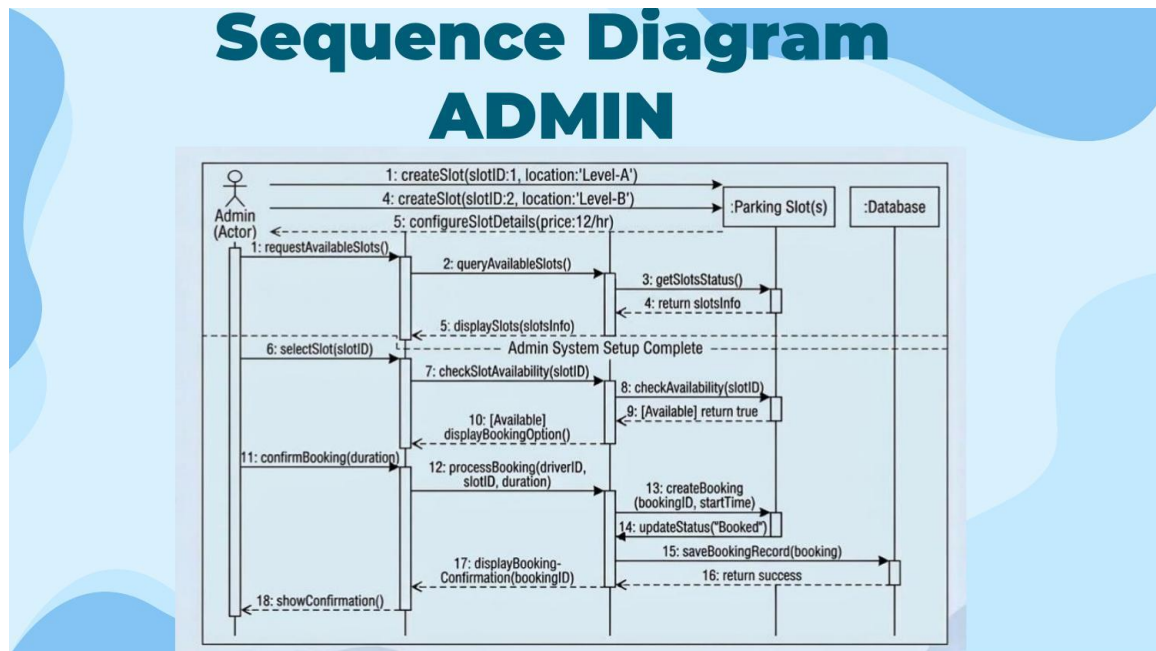


Figure 4: Sequence Diagram - Admin Management Flow

Activity Diagram - Driver

The Driver Activity Diagram starts at login, then proceeds to viewing Available Parking Slots and checking slot availability. If not available, the system shows a message and returns to the view. If available, Confirm Booking triggers three parallel activities: Save Booking Record, Update Parking Slot Status, and Send Booking Confirmation. After all complete, the system displays the confirmation.

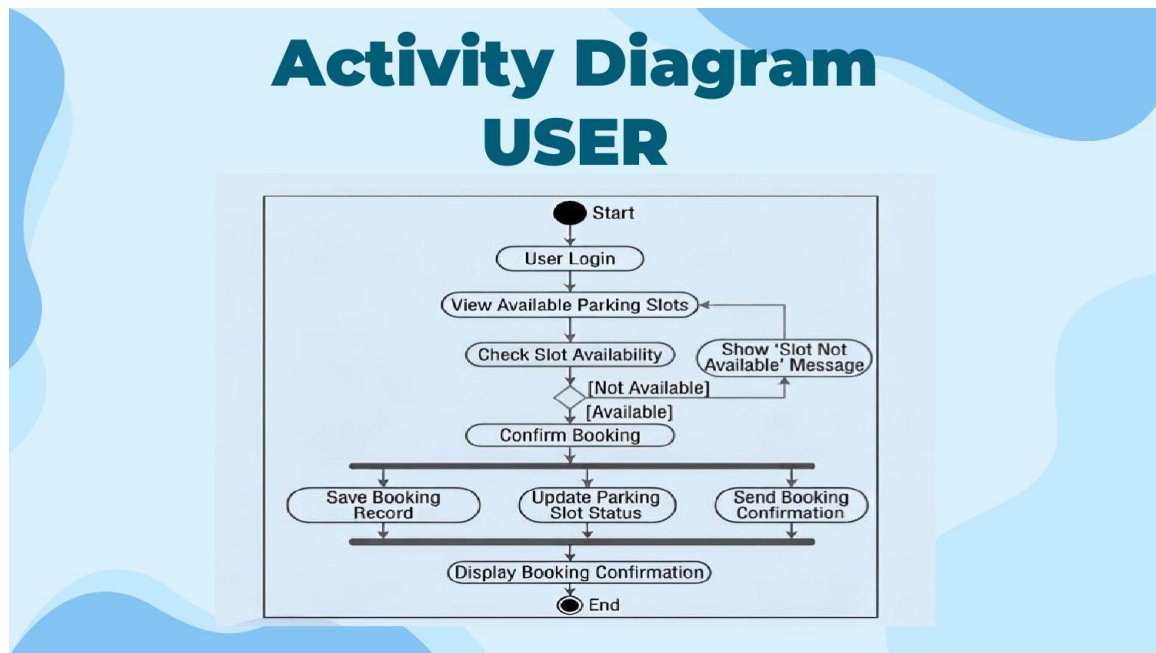


Figure 5: Activity Diagram - Driver Flow

Activity Diagram - Admin

The Admin Activity Diagram starts at login then reaches the Admin Main Menu. Four branches: (1) Add Slot: enter details, save to database, display success. (2) Update Slot: view all, select, edit, save, display success. (3) Delete Slot: view all, select, confirm deletion, delete from database, display success. (4) View All Bookings: retrieve and display a detailed booking report. All paths converge at the End node.

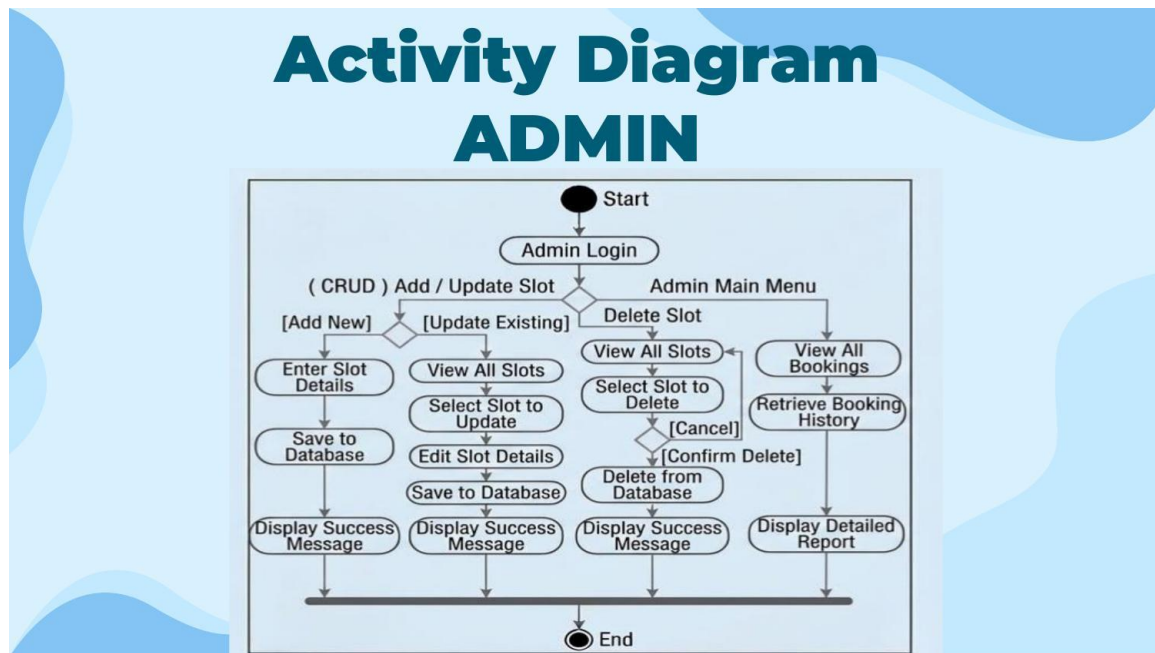


Figure 6: Activity Diagram - Admin Flow

Implementation

The Smart Parking System was implemented as a C# Windows Forms desktop application targeting .NET 9.0 with SQLite as the embedded database. Below are screenshots of all implemented screens.

Login Screen

The Login Screen provides username, password (with show/hide toggle), and role selector. Validation is performed before querying the database.

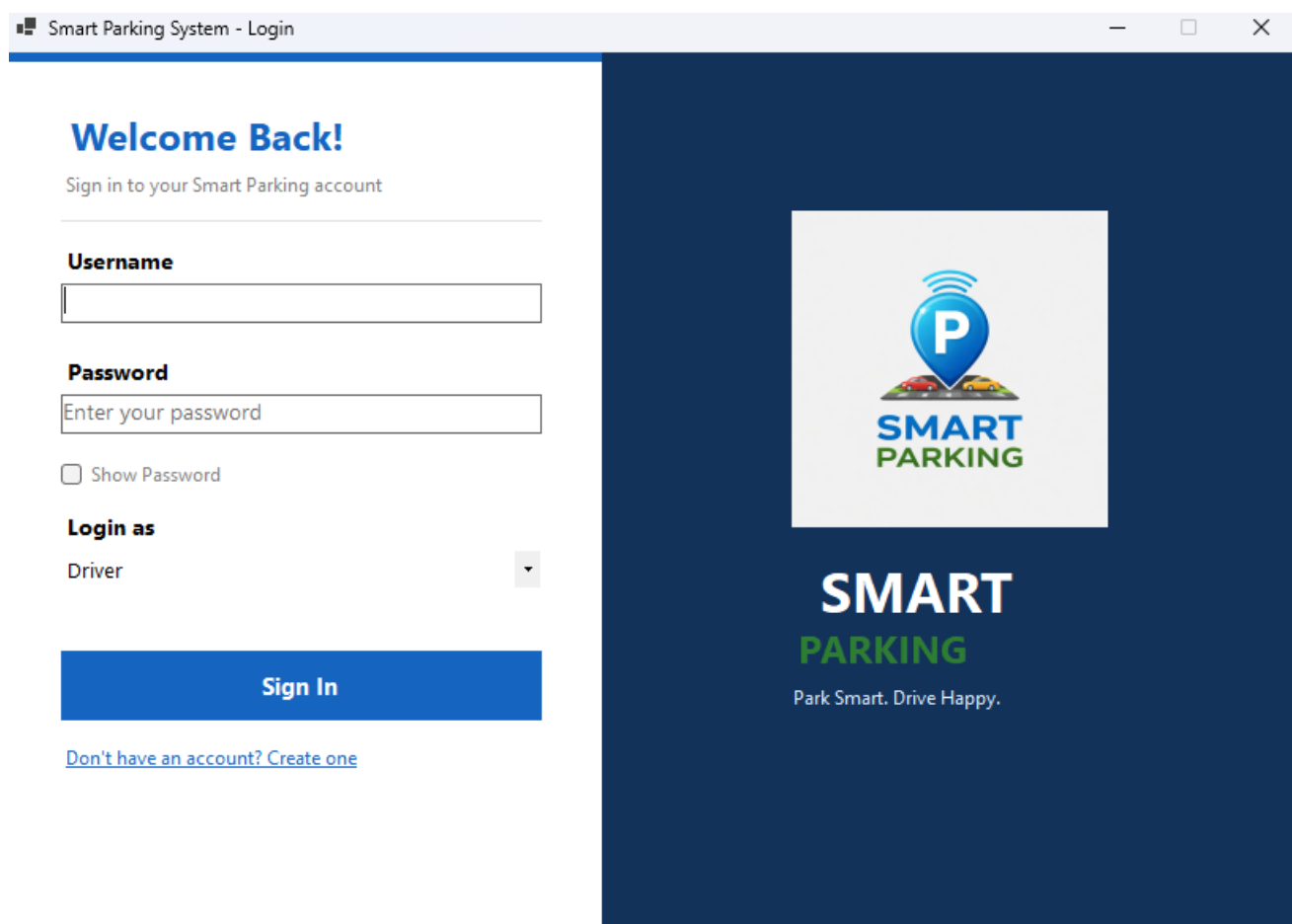
A screenshot of the 'Smart Parking System - Login' window. The window has a title bar with standard Windows controls. The main area is split into two panels. The left panel is white and contains the login form. It starts with 'Welcome Back!' in blue, followed by 'Sign in to your Smart Parking account'. There are input fields for 'Username' and 'Password' (with placeholder 'Enter your password'). A 'Show Password' checkbox is below the password field. A 'Login as' dropdown menu is set to 'Driver'. A blue 'Sign In' button is at the bottom of the form, with a link 'Don't have an account? Create one' below it. The right panel is dark blue and features the 'SMART PARKING' logo (a blue 'P' with a signal icon) and the text 'SMART PARKING' in large white and green letters, with the tagline 'Park Smart. Drive Happy.' below it.

Figure 7: Login Screen

Registration Screen

The Registration Screen allows new users to create an account by entering full name, unique username, password, and role. Username uniqueness is enforced.

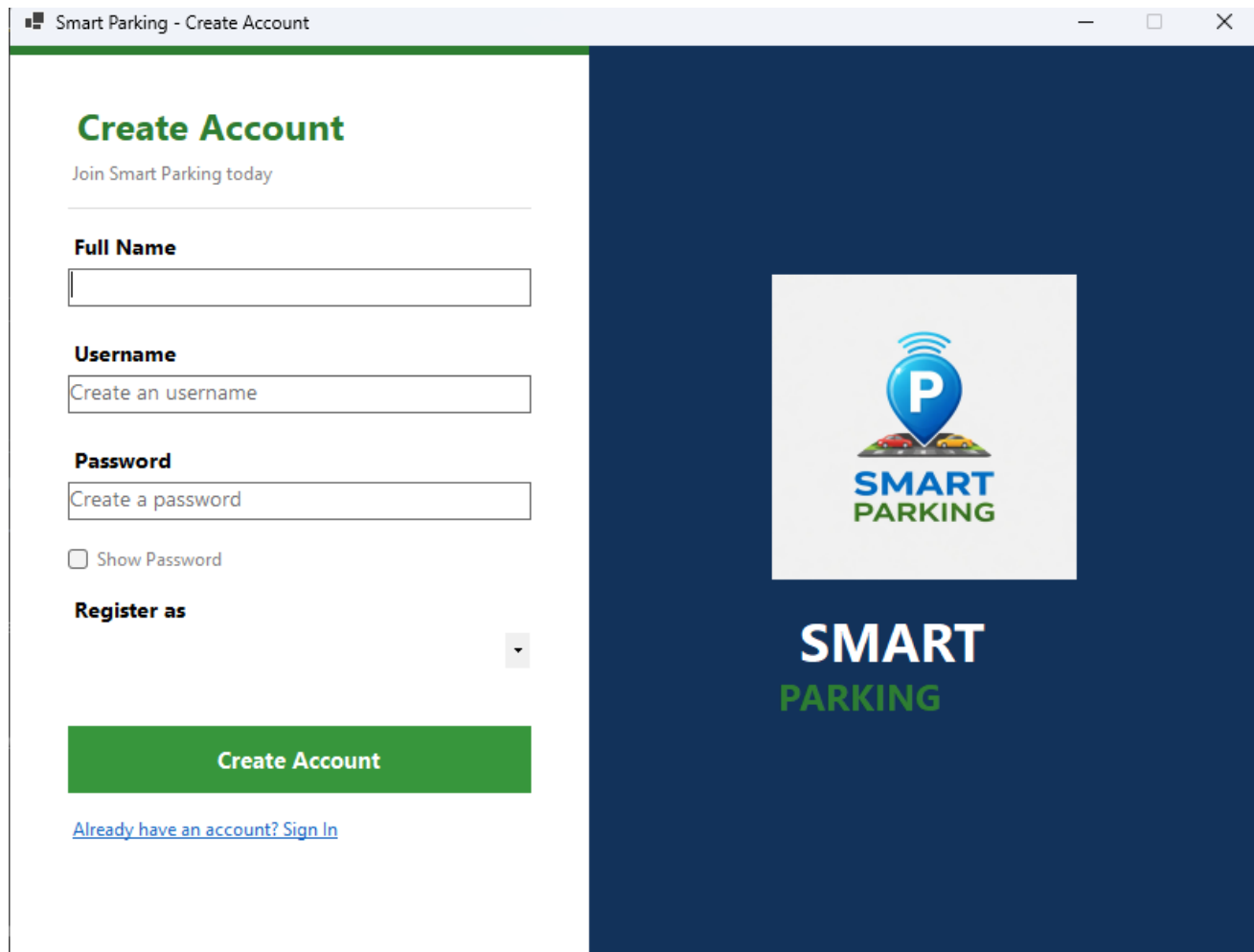
A screenshot of a web browser window titled 'Smart Parking - Create Account'. The page has a dark blue background on the right and a white form area on the left. The form contains the following elements: a green heading 'Create Account', a subtext 'Join Smart Parking today', input fields for 'Full Name', 'Username' (with placeholder 'Create an username'), and 'Password' (with placeholder 'Create a password'), a 'Show Password' checkbox, a 'Register as' dropdown menu, a green 'Create Account' button, and a link 'Already have an account? Sign In'. On the right side of the page, there is a large graphic with a blue 'P' parking sign icon, a car, and the text 'SMART PARKING'.

Figure 8: Registration Screen

Driver Dashboard - Available Slots

The Available Slots tab shows a real-time grid of all available parking slots with location, type, price per hour, status, and a Book Now action button. Three stat cards display Total Slots (10), Available Now (9), and My Bookings (1).


SMART PARKING

Mustafa Hany
Driver

Dashboard
Available Slots
My Bookings
History

Logout

Welcome, Mustafa Hany!

Mon, 11 May 2026

Total Slots
10

Available Now
9

My Bookings
1

Available Slots
My Bookings
History

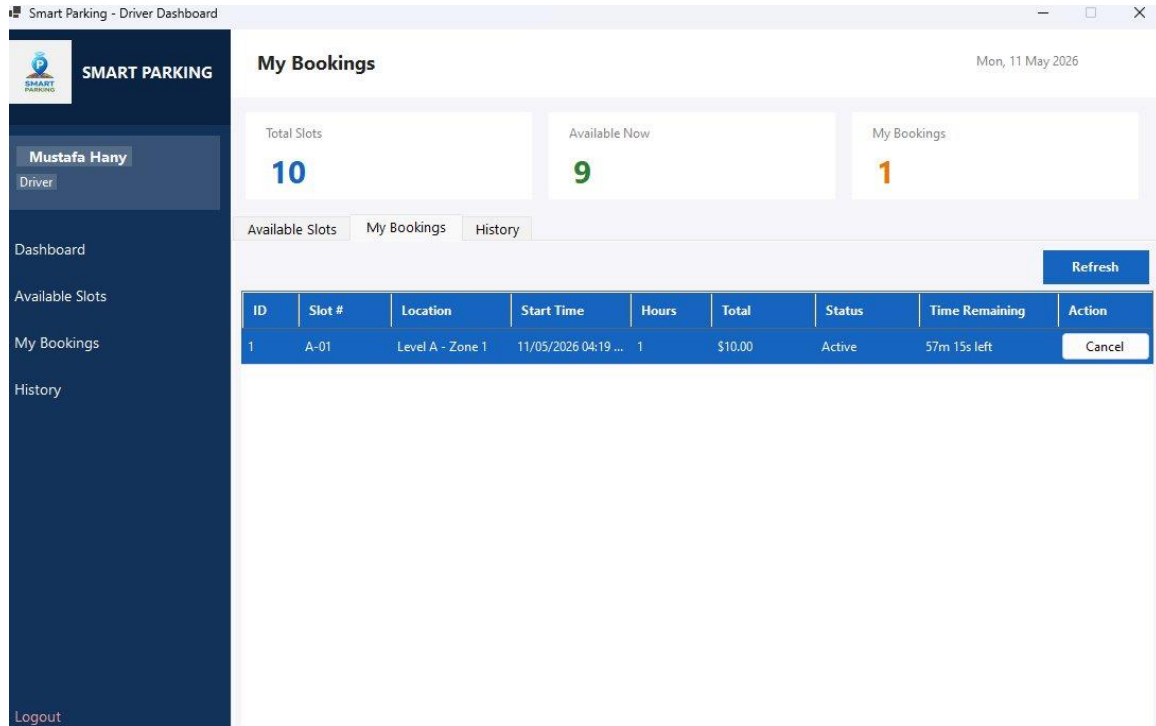
Refresh

ID	Slot #	Location	Type	Price/Hr	Status	Action
2	A-02	Level A - Zone 1	Regular	10	☒ Available	Book Now
3	A-03	Level A - Zone 1	Regular	10	☒ Available	Book Now
4	A-04	Level A - Zone 2	VIP	20	☒ Available	Book Now
5	A-05	Level A - Zone 2	VIP	20	☒ Available	Book Now
6	B-01	Level B - Zone 1	Regular	10	☒ Available	Book Now
7	B-02	Level B - Zone 1	Regular	10	☒ Available	Book Now
8	B-03	Level B - Zone 2	Disabled	5	☒ Available	Book Now
9	B-04	Level B - Zone 2	Regular	10	☒ Available	Book Now
10	B-05	Level B - Zone 2	Regular	10	☒ Available	Book Now

Figure 9: Driver Dashboard - Available Slots Tab

Driver Dashboard - My Bookings

The My Bookings tab displays active bookings with a real-time Time Remaining countdown (e.g., 57m 15s left) that updates every second, and a Cancel button for immediate cancellation.



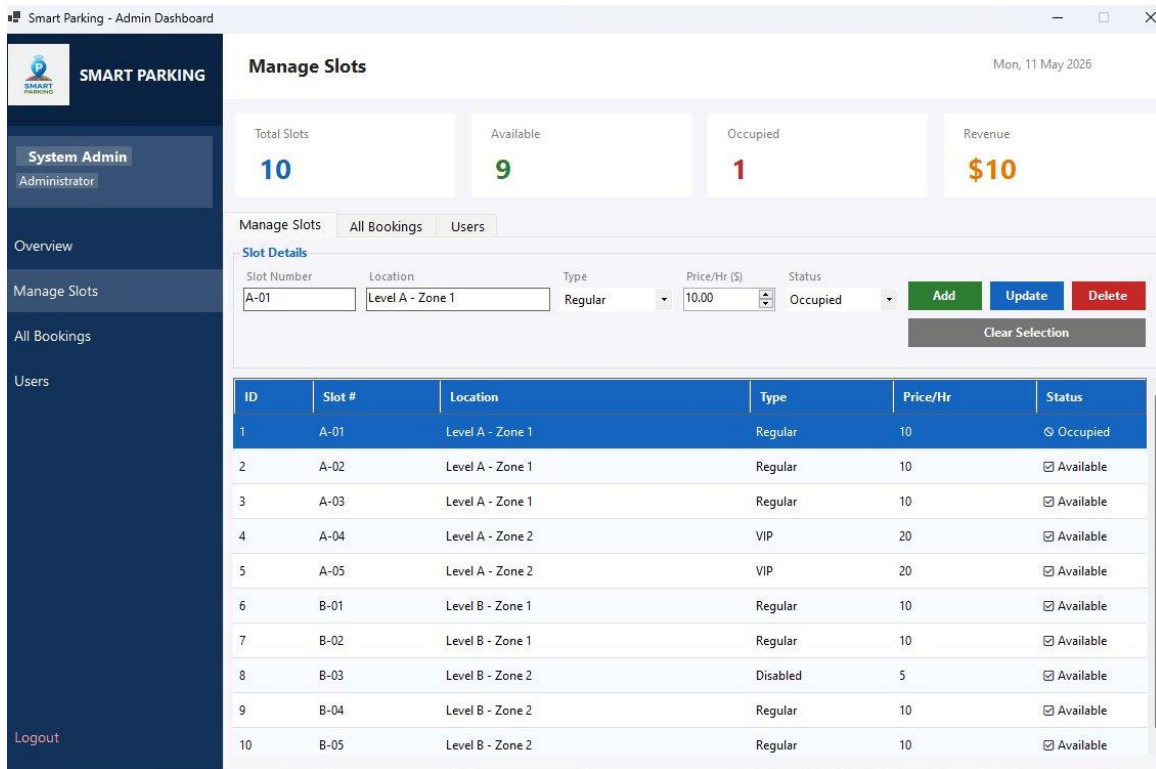
The screenshot shows the 'Smart Parking - Driver Dashboard' interface. The left sidebar contains the user's name 'Mustafa Hany' and role 'Driver', along with navigation links for Dashboard, Available Slots, My Bookings (selected), and History, and a Logout button. The main content area is titled 'My Bookings' and shows the date 'Mon, 11 May 2026'. It features three summary cards: 'Total Slots' (10), 'Available Now' (9), and 'My Bookings' (1). Below these are tabs for 'Available Slots', 'My Bookings' (selected), and 'History'. A 'Refresh' button is located to the right of the tabs. The 'My Bookings' tab displays a table with one active booking:

ID	Slot #	Location	Start Time	Hours	Total	Status	Time Remaining	Action
1	A-01	Level A - Zone 1	11/05/2026 04:19 ...	1	\$10.00	Active	57m 15s left	Cancel

Figure 10: Driver Dashboard - My Bookings Tab with Live Countdown

Admin Dashboard - Manage Slots

The Manage Slots tab includes an inline form for Add, Update, Delete, and Clear Selection operations, plus a complete grid of all parking slots with their current status.



Smart Parking - Admin Dashboard Mon, 11 May 2026

SMART PARKING

System Admin
Administrator

Overview

Manage Slots

All Bookings Users

Slot Details

Slot Number: A-01 Location: Level A - Zone 1 Type: Regular Price/Hr (\$): 10.00 Status: Occupied

Add **Update** **Delete**

Clear Selection

ID	Slot #	Location	Type	Price/Hr	Status
1	A-01	Level A - Zone 1	Regular	10	⊗ Occupied
2	A-02	Level A - Zone 1	Regular	10	☑ Available
3	A-03	Level A - Zone 1	Regular	10	☑ Available
4	A-04	Level A - Zone 2	VIP	20	☑ Available
5	A-05	Level A - Zone 2	VIP	20	☑ Available
6	B-01	Level B - Zone 1	Regular	10	☑ Available
7	B-02	Level B - Zone 1	Regular	10	☑ Available
8	B-03	Level B - Zone 2	Disabled	5	☑ Available
9	B-04	Level B - Zone 2	Regular	10	☑ Available
10	B-05	Level B - Zone 2	Regular	10	☑ Available

Logout

Figure 11: Admin Dashboard - Manage Slots Tab

Admin Dashboard - All Bookings

The All Bookings tab provides a system-wide view of every driver booking with booking ID, driver username, slot, location, start time, hours, total, and status.

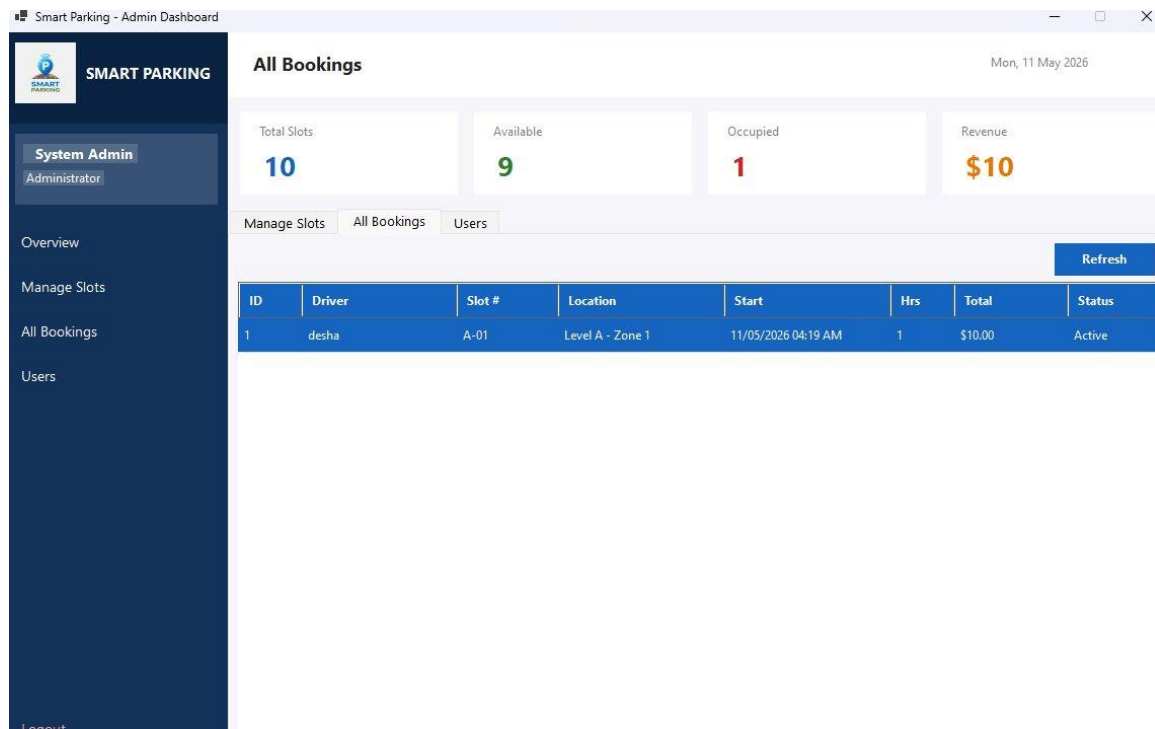
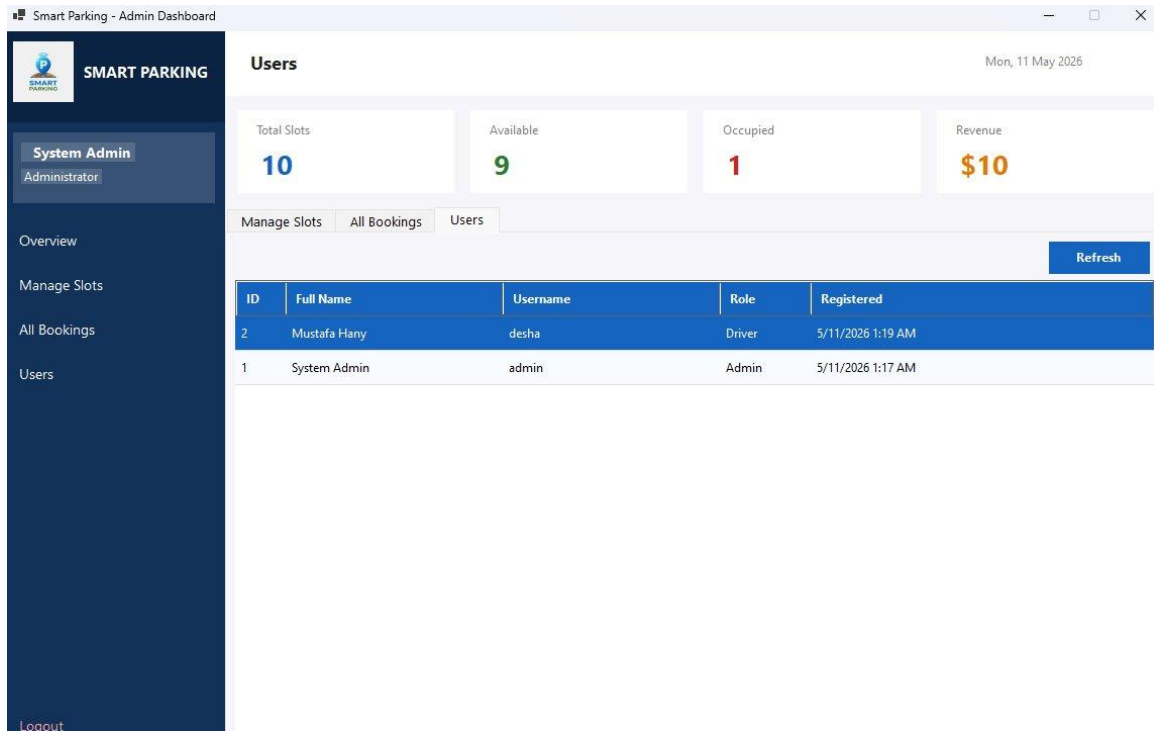


Figure 12: Admin Dashboard - All Bookings Tab

Admin Dashboard - Users

The Users tab lists all registered users with their ID, full name, username, role, and registration date, allowing the administrator to monitor all accounts.



Smart Parking - Admin Dashboard

SMART PARKING

System Admin
Administrator

Overview
Manage Slots
All Bookings
Users

Logout

Users

Mon, 11 May 2026

Total Slots: 10
Available: 9
Occupied: 1
Revenue: \$10

Manage Slots All Bookings Users

Refresh

ID	Full Name	Username	Role	Registered
2	Mustafa Hany	desha	Driver	5/11/2026 1:19 AM
1	System Admin	admin	Admin	5/11/2026 1:17 AM

Figure 13: Admin Dashboard - Users Tab



Testing

The Smart Parking System was tested using black-box functional testing methodology. Test cases cover all functional requirements: authentication, slot operations, booking lifecycle (creation, expiry, cancellation), and administrative functions.

TC-01: Valid Driver Login - Input: Username: desha, Password: ***, Role: Driver - Result: PASS

TC-02: Valid Admin Login - Input: Username: admin, Password: admin123, Role: Admin - Result: PASS

TC-03: Invalid Credentials - Input: Wrong username/password - Expected: Error message shown - Result: PASS

TC-04: Register New Account - Input: Name, Username, Password - Expected: Account created - Result: PASS

TC-05: Duplicate Username - Input: Existing username - Expected: Error shown - Result: PASS

TC-06: Empty Fields Validation - Input: Empty username, click Sign In - Expected: Validation error - Result: PASS

TC-07: View Available Slots - Driver opens Available Slots tab - Expected: Grid shows all available slots - Result: PASS

TC-08: Book a Slot - Driver clicks Book Now on A-01, Duration: 1hr - Expected: Slot Occupied, booking in My Bookings - Result: PASS

TC-09: Time Remaining Display - Driver opens My Bookings - Expected: Countdown visible (57m 15s left) - Result: PASS

TC-10: Booking Auto-Expiry - Wait until EndTime passes - Expected: Slot auto-released, Status: Completed - Result: PASS

TC-11: Cancel Booking - Driver clicks Cancel on active booking - Expected: Booking cancelled, slot freed - Result: PASS

TC-12: View Booking History - Driver opens History tab - Expected: All bookings listed - Result: PASS

TC-13: Admin Add Slot - Admin enters C-01, Level C, clicks Add - Expected: Slot added to grid - Result: PASS

TC-14: Admin Update Slot - Admin selects A-01, changes price, clicks Update - Expected: Price updated - Result: PASS

TC-15: Admin Delete Slot - Admin selects slot, clicks Delete, confirms - Expected: Slot removed - Result: PASS

TC-16: View All Bookings (Admin) - Admin opens All Bookings tab - Expected: All driver bookings shown - Result: PASS

TC-17: View Users (Admin) - Admin opens Users tab - Expected: All users listed with roles - Result: PASS



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